



Figure 9-8: Calling `turtle.begin_fill()`, drawing a path, and calling `turtle.end_fill()` creates a filled-in shape.

The base case is reached when the rectangles of the 3×3 become smaller than six steps on a side. You can change the `MIN_SIZE` constant to a larger value to make the program reach the base case sooner. The source code for *sierpinskiCarpet.py* is as follows:

```
import turtle
turtle.tracer(10, 0) # Increase the first argument to speed up the drawing.
turtle.setworldcoordinates(0, 0, 700, 700)
turtle.hideturtle()

MIN_SIZE = 6 # Try changing this to decrease/increase the amount of recursion.
DRAW_SOLID = True

def isTooSmall(width, height):
    # Determine if the rectangle is too small to draw.
    return width < MIN_SIZE or height < MIN_SIZE

def drawCarpet(x, y, width, height):
    # The x and y are the lower-left corner of the carpet.

    # Move the pen into position.
    turtle.penup()
    turtle.goto(x, y)

    # Draw the outer rectangle.
    turtle.pendown()
    if DRAW_SOLID:
        turtle.fillcolor('black')
        turtle.begin_fill()
    turtle.goto(x, y + height)
    turtle.goto(x + width, y + height)
    turtle.goto(x + width, y)
    turtle.goto(x, y)
    if DRAW_SOLID:
```